Hosting Website using Public VM but server is on Private VM

ARM Template

```
{
  "$schema":
"https://schema.management.azure.com/schemas/2019-04-01/deploymentTempl
ate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "adminUsername": { "type": "string" },
    "adminPassword": { "type": "secureString" },
    "location": { "type": "string", "defaultValue": "eastus2" }
  },
  "variables": {
    "vnetName": "prod-vnet",
    "publicSubnet": "public-subnet",
    "privateSubnet": "private-subnet",
    "publicVMName": "public-vm",
    "privateVMName": "private-vm",
    "publicIPName": "public-ip",
    "publicNicName": "public-nic",
    "privateNicName": "private-nic",
    "publicNSGName": "public-nsg",
    "privateNSGName": "private-nsg",
    "privateVMStaticIP": "10.0.2.4"
  },
  "resources": [
    {
      "type": "Microsoft.Network/networkSecurityGroups",
      "apiVersion": "2023-02-01",
      "name": "[variables('publicNSGName')]",
      "location": "[parameters('location')]",
      "properties": {
        "securityRules": [
          {
            "name": "Allow-HTTP",
            "properties": {
              "priority": 100,
              "direction": "Inbound",
              "access": "Allow",
              "protocol": "Tcp",
```

```
            "sourceAddressPrefix": "*",
            "destinationPortRange": "80",
            "sourcePortRange": "*",
            "destinationAddressPrefix": "*"
          }
        },
        {
          "name": "Allow-SSH",
          "properties": {
            "priority": 110,
            "direction": "Inbound",
            "access": "Allow",
            "protocol": "Tcp",
            "sourceAddressPrefix": "*",
            "destinationPortRange": "22",
            "sourcePortRange": "*",
            "destinationAddressPrefix": "*"
          }
        }
      ]
    }
  },
  {
    "type": "Microsoft.Network/networkSecurityGroups",
    "apiVersion": "2023-02-01",
    "name": "[variables('privateNSGName')]",
    "location": "[parameters('location')]",
    "properties": {
      "securityRules": [
        {
          "name": "Allow-HTTP-From-Public",
          "properties": {
            "priority": 100,
            "direction": "Inbound",
            "access": "Allow",
            "protocol": "Tcp",
            "sourceAddressPrefix": "10.0.1.0/24",
            "destinationPortRange": "80",
            "sourcePortRange": "*",
            "destinationAddressPrefix": "*"
          }
        }
      ]
```

```
      }
    },
    {
      "type": "Microsoft.Network/virtualNetworks",
      "apiVersion": "2023-02-01",
      "name": "[variables('vnetName')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Network/networkSecurityGroups',
variables('publicNSGName'))]",
        "[resourceId('Microsoft.Network/networkSecurityGroups',
variables('privateNSGName'))]"
      ],
      "properties": {
        "addressSpace": { "addressPrefixes": ["10.0.0.0/16"] },
        "subnets": [
          {
            "name": "[variables('publicSubnet')]",
            "properties": {
              "addressPrefix": "10.0.1.0/24",
              "networkSecurityGroup": {
                "id":
"[resourceId('Microsoft.Network/networkSecurityGroups',
variables('publicNSGName'))]"
              }
            }
          },
          {
            "name": "[variables('privateSubnet')]",
            "properties": {
              "addressPrefix": "10.0.2.0/24",
              "networkSecurityGroup": {
                "id":
"[resourceId('Microsoft.Network/networkSecurityGroups',
variables('privateNSGName'))]"
              }
            }
          }
        ]
      }
    },
    {
      "type": "Microsoft.Network/publicIPAddresses",
```

```
      "apiVersion": "2023-02-01",
      "name": "[variables('publicIPName')]",
      "location": "[parameters('location')]",
      "sku": { "name": "Standard" },
      "properties": { "publicIPAllocationMethod": "Static" }
    },
    {
      "type": "Microsoft.Network/networkInterfaces",
      "apiVersion": "2023-02-01",
      "name": "[variables('publicNicName')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Network/publicIPAddresses',
variables('publicIPName'))]",
        "[resourceId('Microsoft.Network/virtualNetworks',
variables('vnetName'))]"
      ],
      "properties": {
        "ipConfigurations": [{
          "name": "ipconfig",
          "properties": {
            "subnet": {
              "id":
"[resourceId('Microsoft.Network/virtualNetworks/subnets',
variables('vnetName'), variables('publicSubnet'))]"
            },
            "privateIPAllocationMethod": "Dynamic",
            "publicIPAddress": {
              "id": "[resourceId('Microsoft.Network/publicIPAddresses',
variables('publicIPName'))]"
            }
          }
        }]
      }
    },
    {
      "type": "Microsoft.Network/networkInterfaces",
      "apiVersion": "2023-02-01",
      "name": "[variables('privateNicName')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Network/virtualNetworks',
variables('vnetName'))]"
```

```
      ],
      "properties": {
        "ipConfigurations": [{
          "name": "ipconfig",
          "properties": {
            "subnet": {
              "id":
"[resourceId('Microsoft.Network/virtualNetworks/subnets',
variables('vnetName'), variables('privateSubnet'))]"
            },
            "privateIPAllocationMethod": "Static",
            "privateIPAddress": "[variables('privateVMStaticIP')]"
          }
        }]
      }
    },
    {
      "type": "Microsoft.Compute/virtualMachines",
      "apiVersion": "2023-03-01",
      "name": "[variables('privateVMName')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Network/networkInterfaces',
variables('privateNicName'))]"
      ],
      "properties": {
        "hardwareProfile": { "vmSize": "Standard_D2s_v3" },
        "osProfile": {
          "computerName": "private-vm",
          "adminUsername": "[parameters('adminUsername')]",
          "adminPassword": "[parameters('adminPassword')]",
          "linuxConfiguration": { "disablePasswordAuthentication":
false }
        },
        "storageProfile": {
          "imageReference": {
            "publisher": "Canonical",
            "offer": "0001-com-ubuntu-server-jammy",
            "sku": "22_04-lts",
            "version": "latest"
          },
          "osDisk": { "createOption": "FromImage" }
        },
```

```
        "networkProfile": {
          "networkInterfaces": [{
            "id": "[resourceId('Microsoft.Network/networkInterfaces',
variables('privateNicName'))]"
          }]
        }
      }
    },
    {
      "type": "Microsoft.Compute/virtualMachines/extensions",
      "apiVersion": "2022-11-01",
      "name": "[concat(variables('privateVMName'), '/installWeb')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Compute/virtualMachines',
variables('privateVMName'))]"
      ],
      "properties": {
        "publisher": "Microsoft.Azure.Extensions",
        "type": "CustomScript",
        "typeHandlerVersion": "2.1",
        "settings": {
          "commandToExecute": "bash -c 'export
DEBIAN_FRONTEND=noninteractive && sudo apt update -y && sudo apt
install -y nginx && echo \"<h1>Private VM App Server</h1>\" | sudo tee
/var/www/html/index.html && sudo systemctl enable nginx && sudo
systemctl restart nginx'"
        }
      }
    },
    {
      "type": "Microsoft.Compute/virtualMachines",
      "apiVersion": "2023-03-01",
      "name": "[variables('publicVMName')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Network/networkInterfaces',
variables('publicNicName'))]"
      ],
      "properties": {
        "hardwareProfile": { "vmSize": "Standard_D2s_v3" },
        "osProfile": {
          "computerName": "public-vm",
```

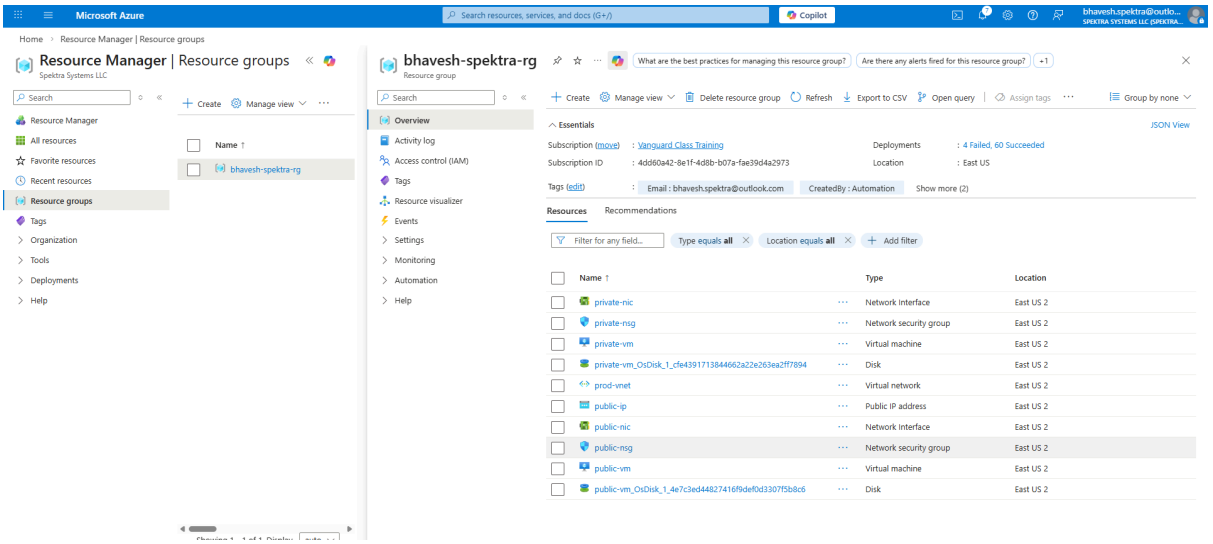
```
          "adminUsername": "[parameters('adminUsername')]",
          "adminPassword": "[parameters('adminPassword')]",
          "linuxConfiguration": { "disablePasswordAuthentication":
false }
        },
        "storageProfile": {
          "imageReference": {
            "publisher": "Canonical",
            "offer": "0001-com-ubuntu-server-jammy",
            "sku": "22_04-lts",
            "version": "latest"
          },
          "osDisk": { "createOption": "FromImage" }
        },
        "networkProfile": {
          "networkInterfaces": [{
            "id": "[resourceId('Microsoft.Network/networkInterfaces',
variables('publicNicName'))]"
          }]
        }
      }
    },
    {
      "type": "Microsoft.Compute/virtualMachines/extensions",
      "apiVersion": "2022-11-01",
      "name": "[concat(variables('publicVMName'), '/installProxy')]",
      "location": "[parameters('location')]",
      "dependsOn": [
        "[resourceId('Microsoft.Compute/virtualMachines',
variables('publicVMName'))]",
        "[resourceId('Microsoft.Compute/virtualMachines/extensions',
variables('privateVMName'), 'installWeb')]"
      ],
      "properties": {
        "publisher": "Microsoft.Azure.Extensions",
        "type": "CustomScript",
        "typeHandlerVersion": "2.1",
        "settings": {
          "commandToExecute": "bash -c 'export
DEBIAN_FRONTEND=noninteractive && sudo apt update -y && sudo apt
install -y nginx && echo \"server { listen 80; location / { proxy_pass
http://10.0.2.4; proxy_set_header Host \\$host; proxy_set_header
X-Real-IP \\$remote_addr; } }\" | sudo tee
```

```
/etc/nginx/sites-available/default && sudo nginx -t && sudo systemctl
restart nginx'"
        }
      }
    }
  ],
  "outputs": {
    "PublicIPAddress": {
      "type": "string",
      "value":
"[reference(resourceId('Microsoft.Network/publicIPAddresses',
variables('publicIPName'))).ipAddress]"
    }
  }
}
```

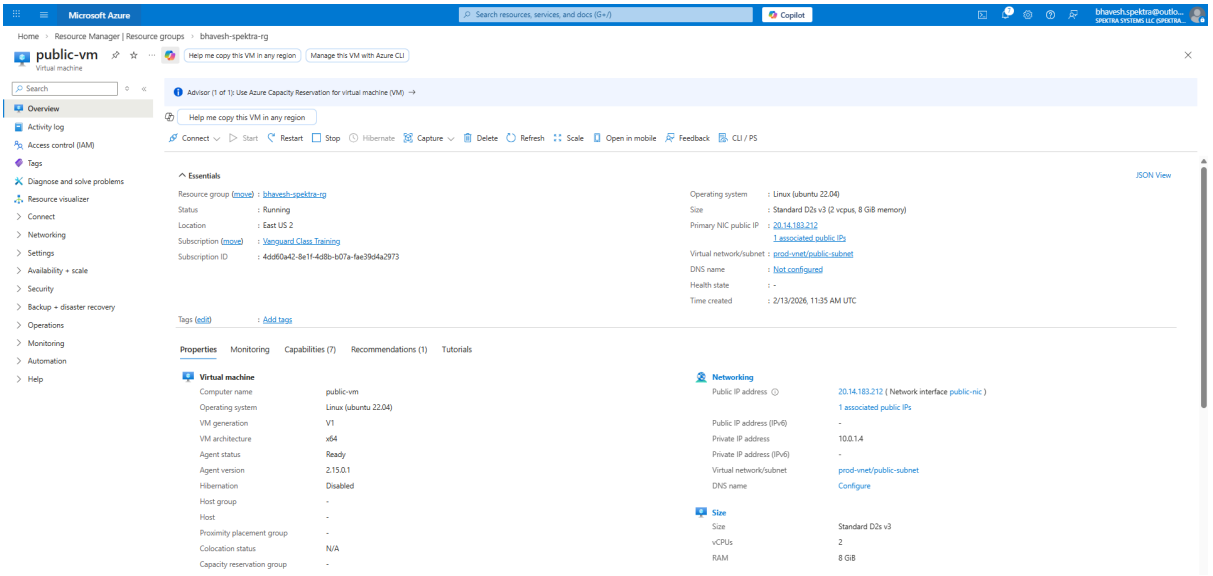
Parameter File

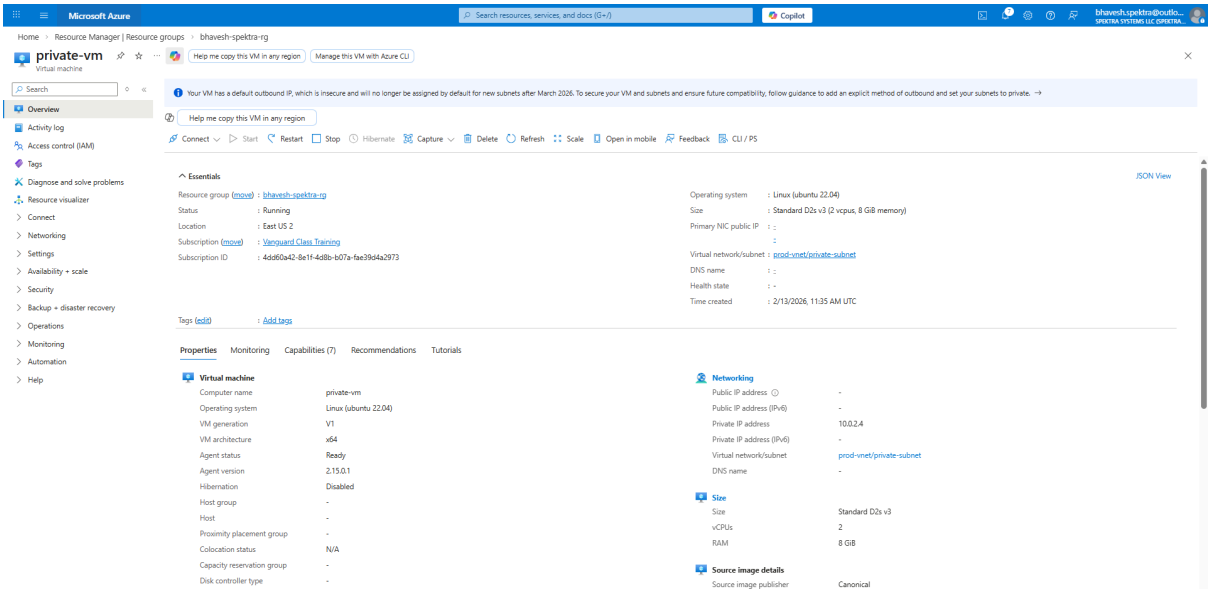
```
{
  "$schema":
"https://schema.management.azure.com/schemas/2019-04-01/deploymentParam
eters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "adminUsername": {
      "value": "azureuser"
    },
    "adminPassword": {
      "value": "YourStrongP@ssword123!"
    },
    "location": {
      "value": "eastus2"
    }
  }
}
```

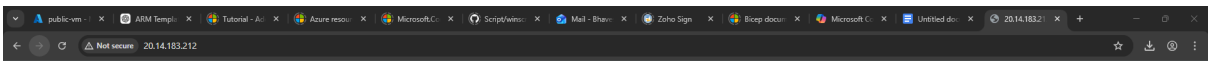
Resources


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
bhavesh.spektra@outlo...
SPEKTRA SYSTEMS LLC (SPEKTRA...
Home > Resource Manager | Resource groups
Resource Manager | Resource groups
Spektra Systems LLC
Search
Resource Manager
All resources
Favorite resources
Recent resources
Resource groups
Tags
Organization
Tools
Deployments
Help
Create
Manage view
Name ↑
bhavesh-spektra-rg
bhavesh-spektra-rg
Resource group
What are the best practices for managing this resource group?
Are there any alerts fired for this resource group?
+1
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Monitoring
Automation
Help
Delete resource group
Refresh
Export to CSV
Open query
Assign tags
Group by none
Essentials
JSON View
Subscription (move) : Vanguard Class Training
Subscription ID : 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973
Deployments : 4 Failed, 60 Succeeded
Location : East US
Tags (edit) : Email : bhavesh.spektra@outlook.com
CreatedBy : Automation
Show more (2)
Resources
Recommendations
Filter for any field...
Type equals all
Location equals all
Add filter
Name ↑
Type
Location
private-nic
Network Interface
East US 2
private-nsg
Network security group
East US 2
private-vm
Virtual machine
East US 2
private-vm_OsDisk_1_cfe4391713844662a22e263ea2ff7894
Disk
East US 2
prod-vnet
Virtual network
East US 2
public-ip
Public IP address
East US 2
public-nic
Network Interface
East US 2
public-nsg
Network security group
East US 2
public-vm
Virtual machine
East US 2
public-vm_OsDisk_1_4e7c3ed44827416f9def0d3307f5b8c6
Disk
East US 2

private-vm
Virtual machine
private-vm_OsDisk_1_cfe4391
713844602a22a263a52ff7894
Disk
private-nic
Network interface
prod-vnet
Virtual network
public-nic
Network interface
private-nsg
Network security group
public-nsg
Network security group
public-vm
Virtual machine
public-ip
Public IP address
public-vm_OsDisk_1_4e7c3ed
448274 1bf9dafb0d330ff5b8c6
Disk







Private VM App Server

```
PS E:\Internship\Bicep> ssh azureuser@20.14.183.212
The authenticity of host '20.14.183.212 (20.14.183.212)' can't be established.
ED25519 key fingerprint is SHA256:oolEsjRX2vbrnYY1Fmd5hGEf8wm6Wcn0rOiZAifAjcU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.14.183.212' (ED25519) to the list of known hosts.
azureuser@20.14.183.212's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1044-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Feb 13 11:38:21 UTC 2026

  System load:  0.41               Processes:             132
  Usage of /:   6.4% of 28.89GB    Users logged in:       0
  Memory usage: 4%                 IPv4 address for eth0: 10.0.1.4
  Swap usage:   0%
```

```
azureuser@public-vm:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-02-13 11:37:23 UTC; 1min 25s ago
       Docs: man:nginx(8)
    Process: 2451 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 2452 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2453 (nginx)
      Tasks: 3 (limit: 9524)
     Memory: 3.4M
        CPU: 27ms
     CGroup: /system.slice/nginx.service
             ├─2453 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─2454 "nginx: worker process" "" "" "
```

```
azureuser@private-vm:~$ sudo apt update -y
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
azureuser@private-vm:~$ sudo apt install -y nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.18.0-6ubuntu14.8).
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
azureuser@private-vm:~$ sudo apt install -y apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support
  ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 17 not upgraded.
Need to get 2153 kB of archives.
After this operation, 8556 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9170 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0 amd64 5.3.6-1build1 [140 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.18 [1360 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.18 [165 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.18 [89.5 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all 3.70+nmu1ubuntu1 [23.8 kB]
```

```
azureuser@private-vm:~$ sudo systemctl status apache2
× apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: failed (Result: exit-code) since Fri 2026-02-13 11:42:04 UTC; 9s ago
       Docs: https://httpd.apache.org/docs/2.4/
        CPU: 22ms

Feb 13 11:42:04 private-vm systemd[1]: Starting The Apache HTTP Server...
Feb 13 11:42:04 private-vm apachectl[3593]: (98)Address already in use: AH00072: make_sock: could not bind to address [::]:80
Feb 13 11:42:04 private-vm apachectl[3593]: (98)Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80
Feb 13 11:42:04 private-vm apachectl[3593]: no listening sockets available, shutting down
Feb 13 11:42:04 private-vm apachectl[3593]: AH00015: Unable to open logs
Feb 13 11:42:04 private-vm apachectl[3583]: Action 'start' failed.
Feb 13 11:42:04 private-vm apachectl[3583]: The Apache error log may have more information.
Feb 13 11:42:04 private-vm systemd[1]: apache2.service: Control process exited, code=exited, status=1/FAILURE
Feb 13 11:42:04 private-vm systemd[1]: apache2.service: Failed with result 'exit-code'.
Feb 13 11:42:04 private-vm systemd[1]: Failed to start The Apache HTTP Server.
azureuser@private-vm:~$ curl http://localhost
<h1>Private VM App Server</h1>
azureuser@private-vm:~$ sudo systemctl status apache2
× apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: failed (Result: exit-code) since Fri 2026-02-13 11:42:04 UTC; 36s ago
       Docs: https://httpd.apache.org/docs/2.4/
        CPU: 22ms

Feb 13 11:42:04 private-vm systemd[1]: Starting The Apache HTTP Server...
Feb 13 11:42:04 private-vm apachectl[3593]: (98)Address already in use: AH00072: make_sock: could not bind to address [::]:80
Feb 13 11:42:04 private-vm apachectl[3593]: (98)Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80
Feb 13 11:42:04 private-vm apachectl[3593]: no listening sockets available, shutting down
Feb 13 11:42:04 private-vm apachectl[3593]: AH00015: Unable to open logs
Feb 13 11:42:04 private-vm apachectl[3583]: Action 'start' failed.
Feb 13 11:42:04 private-vm apachectl[3583]: The Apache error log may have more information.
Feb 13 11:42:04 private-vm systemd[1]: apache2.service: Control process exited, code=exited, status=1/FAILURE
Feb 13 11:42:04 private-vm systemd[1]: apache2.service: Failed with result 'exit-code'.
Feb 13 11:42:04 private-vm systemd[1]: Failed to start The Apache HTTP Server.
azureuser@private-vm:~$ sudo apt install lighttpd
Reading package lists... Done
```

```
azureuser@private-vm:~$ sudo apt install lighttpd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  lighttpd-mod-deflate lighttpd-mod-openssl spawn-fcgi
Suggested packages:
  rrdtool php-cgi php-fpm lighttpd-doc lighttpd-mod-authn-gssapi lighttpd-mod-authn-pam lighttpd-mod-authn-sasl lighttpd-mod-geoip
  lighttpd-mod-maxminddb lighttpd-mod-trigger-b4-dl lighttpd-mod-vhostdb-pgsql lighttpd-mod-webdav lighttpd-modules-dbi lighttpd-modules-ldap
  lighttpd-modules-lua lighttpd-modules-mysql
The following NEW packages will be installed:
  lighttpd lighttpd-mod-deflate lighttpd-mod-openssl spawn-fcgi
0 upgraded, 4 newly installed, 0 to remove and 17 not upgraded.
Need to get 437 kB of archives.
After this operation, 1471 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 lighttpd amd64 1.4.63-1ubuntu3.1 [389 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 lighttpd-mod-deflate amd64 1.4.63-1ubuntu3.1 [11.7 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 lighttpd-mod-openssl amd64 1.4.63-1ubuntu3.1 [21.5 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 spawn-fcgi amd64 1.6.4-2 [14.9 kB]
Fetched 437 kB in 0s (10.4 MB/s)
Selecting previously unselected package lighttpd.
(Reading database ... 63845 files and directories currently installed.)
Preparing to unpack .../lighttpd_1.4.63-1ubuntu3.1_amd64.deb ...
Unpacking lighttpd (1.4.63-1ubuntu3.1) ...
Selecting previously unselected package lighttpd-mod-deflate.
Preparing to unpack .../lighttpd-mod-deflate_1.4.63-1ubuntu3.1_amd64.deb ...
Unpacking lighttpd-mod-deflate (1.4.63-1ubuntu3.1) ...
Selecting previously unselected package lighttpd-mod-openssl.
Preparing to unpack .../lighttpd-mod-openssl_1.4.63-1ubuntu3.1_amd64.deb ...
Unpacking lighttpd-mod-openssl (1.4.63-1ubuntu3.1) ...
Selecting previously unselected package spawn-fcgi.
Preparing to unpack .../spawn-fcgi_1.6.4-2_amd64.deb ...
Unpacking spawn-fcgi (1.6.4-2) ...
Setting up lighttpd (1.4.63-1ubuntu3.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/lighttpd.service → /lib/systemd/system/lighttpd.service.
Setting up lighttpd-mod-openssl (1.4.63-1ubuntu3.1) ...
Setting up spawn-fcgi (1.6.4-2) ...
```